

### **Market Structures and their characteristics**

- Perfect competition is an ideal market structure where there are many firms, freedom of entry into the industry with all firms producing identical products. Firms are price takers and there is perfect information for all firms in the market. The other market structures below are collectively referred to as imperfect competition.
- Monopolistic competition is where there are many firms and freedom of entry into an industry but, unlike perfect competition, firms have some control over the product and its price. Firms are price makers.
- Oligopoly is where there are few firms and they have market power to erect barriers to entry to deter competition from new firms. The products can be wide ranging. Firms are price makers as they have some control over price depending on the market power of competitors.
- Monopoly is where there is a single seller in the market. When there is just one firm producing a good or service for the entire market, this is referred to as a pure monopoly. A monopolistic industry may consist of one firm that dominates the market due to it having a very large market share. Barriers to entry into the market are substantial due to patents and other restrictive methods that prohibit competition. A monopoly is a price maker and there is limited information about how the firm operates.
- A particular form of monopoly is a natural monopoly this is one where long-run average costs are lower when there is just one firm in the industry.

### **Barriers to entry and exit**

The existence of substantial barriers to entry into an industry differentiates oligopoly and monopoly from monopolistic competition and perfect competition. Barriers to entry are a range of obstacles that deter or prevent new firms from entering a market to compete with existing firms. Barriers to entry give firms a degree of market power. Decisions can be made by existing firms without the risk of their market share or price being challenged from outside. The construction and maintenance of these barriers can become part of the firm's behavior. Below are some examples of barriers to entry.

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Where the barriers to entry are strong, the market is likely to be dominated by a few large producers. New firms will only enter if they think that the benefits will be greater than the cost of breaking down the barriers to entry.

The concept of barriers to entry is central to understanding where the models of oligopoly and monopoly fit within the spectrum of competition.

**Legal barriers:** In some countries, it may be impossible for new firms to enter an industry because the economic activity is state-owned or the good is produced under license from the government. This is a legal monopoly usually created to achieve social and political objectives. The economic justification lies in the concept of a natural monopoly, where it is more efficient to have a single producer than to have competing firms, for example in the case of railways. Alternatively, in countries where the former state-owned resources have been privatized and the market has been deregulated, the justification for removing barriers to entry is that the injection of competition will bring additional economic and social benefits through a more efficient allocation of resources.

The production process or products of a firm may be protected by a legal monopoly in the form of a patent, whereby competitors cannot copy a product without the permission of the owner. The principle is to try to guarantee exclusivity to entrepreneurs with original ideas but only for a reasonable period of time. The patent acts as a barrier to accessing technology or information.

**Market barriers:** Advertising and brand names with a high degree of consumer loyalty may prove a difficult obstacle to overcome. This explains why many firms regard their expenditure on advertising and promotion as a form of investment. Existing firms can make entry for new firms more difficult where a firm saturates the market with lots of brands, giving its consumers what appears to be a wide range of choice. Branding gives the firm greater market power because consumers do not see the rival firm's product as a close substitute on account of extensive advertising.

Collaboration between existing producers to develop new products may act as a barrier in that the resources necessary to compete are beyond the means of single new producers.

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Market conditions, such as a fall in demand resulting from recession, can leave producers with surplus productive capacity and this will deter entry.

**Cost barriers:** The high fixed cost or setup cost in activities such as electricity generation, aircraft and car production and pharmaceuticals may deter potential entrants. The barrier here is access to capital. Only very large firms will be able to fund the necessary investment. Research and development costs will represent a high proportion of total costs. High sales over a long period of time are needed before the firm is profitable.

Economies of scale can be a barrier because existing large producers are able to produce at a lower average cost than those just starting up. Economies of scale give large firms an opportunity to cut their price in order to eliminate any high-cost producers. This is the concept of predatory pricing, which can be used to remove any new firms that enter the industry.

It may be possible for existing firms to hide the existence of supernormal profit by limit pricing. This involves firms deliberately setting a low price and temporarily abandoning profit maximization as their objective to deter new entrants. It may be in the interest of all the firms to do this and, if they agree, it becomes a form of collusion.

In activities such as consumer electronics, the pace of product innovation is so rapid that existing firms will be working on the next generation of products while launching the current range. Unless new entrants have original ideas or can exploit a new market segment, they are destined to fail.

**Physical barriers:** Some firms may have monopoly access to raw materials, components or retail outlets, which will make it difficult for new entrants to make an impact. Vertically-integrated manufacturing businesses, where a firm is a specialist in different stages of production, will be protected by the fact that their rivals' costs will be higher.

**Barriers to exit:** If a firm is shutting down, then some costs such as research and development costs cannot usually be recovered. The resources are not easily transferred to other uses. They are regarded as sunk costs and act as a barrier to

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exit from the industry because the capital investment will be lost. It is therefore the risk of entering and the high cost of failure that deters potential entrants and so acts as a barrier to entry.

### **PERFECT COMPETITION**

Under conditions of perfect competition the firm is powerless to exert influence on price. It sees the market price as 'given' that is, established by the forces beyond its control. For example in most countries, the individual farmer has no influence on prices at which he sells his wheat, or beef or milk or vegetables. Any changes in the amounts of these things brought to the market will have negligible effects on their prices, so the firm is a price taker.

The demand curve for the output of the single firm therefore, must be horizontal line at the ruling price; in other words, a perfectly elastic demand curve. No matter how many units the firm sells it cannot change the price. It can sell the entire quantity at the ruling price. If it tries to sell at higher price its demand will drop to zero, and there is obviously no incentive to sell at lower price.

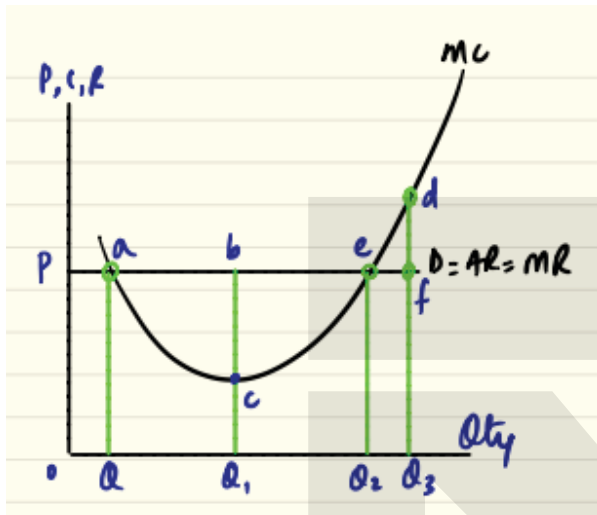
### **FEATURES:**

- Firms are **price takers**: there are so many firms in the industry that each one produces an insignificantly small part of the total industry's supply and therefore has no power to affect the price of the product. It faces a horizontal demand curve at the market price which is determined by the intersection of market demand and supply curves.
- The product that is sold by the firms in the industry is **homogenous**: this means that the product sold by each firm in the industry is a perfect substitute of each other.
- There is complete **freedom of entry** of firms into the industry: existing firms are unable to stop new firms joining the industry. If profits are higher than elsewhere, capital will be freely attracted into that industry.
- Producers and consumers have **perfect knowledge** of the market: that is, producers are fully aware of prices, costs and market opportunities. Consumers are fully aware of the price, quality and availability of the product.

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**THE PROFIT MAXIMIZATION CONDITION:** The firms maximize at a point where  $MC = MR$  i.e. the last unit adds as much to the revenue as it does to the cost. At this point the total profit of the firm has been accumulated or maximized.



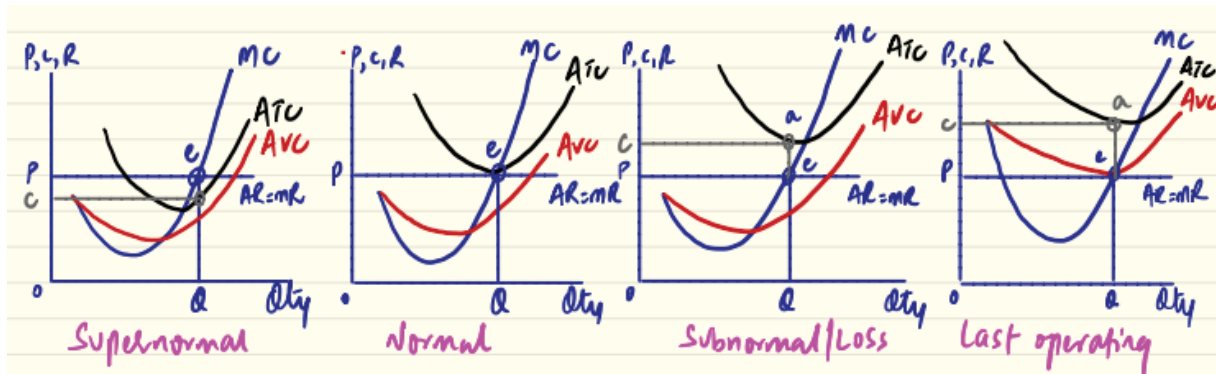
**NORMAL PROFIT:** This is the level of profit that is just enough to persuade firms to stay in the industry in the long run but not enough to attract new firms. In the short run when the firm is earning normal profits, the firms are just covering its total costs. Since the TC also includes the return on capital employed in alternative uses, the firms are earning profits which it could earn in other businesses. So this is enough for the firm to stay in the industry in the long run. In this situation the  $AC=AR$  and thus  $TC=TR$ . The word profit is associated with it is because  $AR > AVC$ .

**SUPERNORMAL OR ABNORMAL PROFITS:** When the firms earn more than normal profits it is said to be earning super normal profits. In this situation  $AR > AC$  is at equilibrium level of output. In the figure the area shaded is the supernormal profits. Firm is covering its ATC, this means AR is covering AVC and AFC.

**SUBNORMAL PROFITS:** In the short run firms may experience subnormal profits (i.e. losses) this mean that firms will be producing where  $AC > AR$ . Still the word profit is associated with it because  $AR > AVC$ .

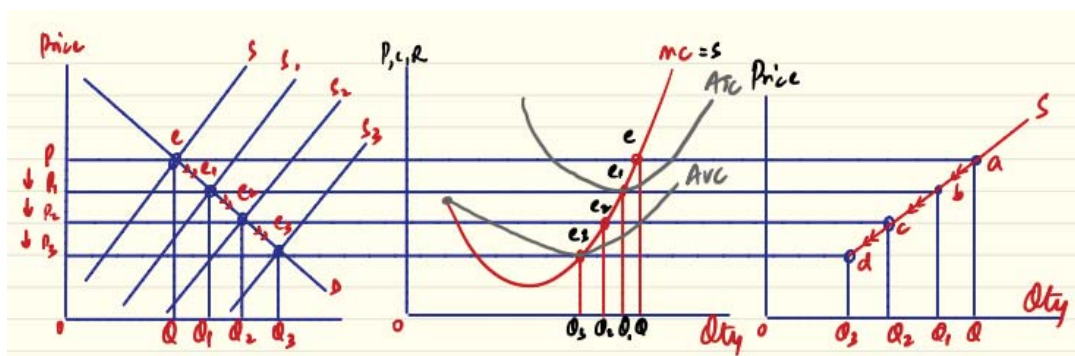
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**LAST OPERATING POINT:** In the short run the firms will continue to produce as long as it is covering VC i.e. till the price falls to the minimum of AVC i.e. last operating point, price falls below that point the firm will shut down in the short run as it will not be covering even the AVC. If the price is somewhere between AVC and AC, the firm is covering the AVC and part of the AFC, that is subnormal profits.

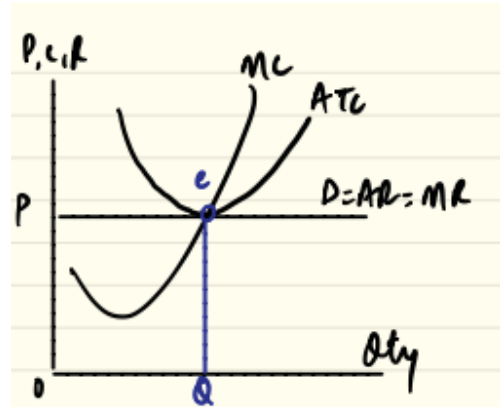
**The supply curve of a firm:** In the short run the equilibrium is determined at a point where  $MC=MR$ . This point on the MC curve shows a particular quantity that will be supplied at the ruling market price, if the price falls (mainly due to a fall in demand), the producer will adjust his output such that MC is equal to the new price or MR. This goes on if the price continues to fall until it reaches the minimum of AVC which is the last point of production in the short-run. In this way we get the series of points along the MC curve which shows a positive relationship between different prices and the quantities which the producer is willing to supply. It can be safely concluded that the short run supply curve of the firm under perfect competition is the part of the marginal cost curve after intersecting the average variable cost.



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**THE LONG RUN EQUILIBRIUM:** In the long run new firms will enter the industry by looking at the supernormal profits of the existing firms. This will increase the supply causing the price to fall until the profits fall to zero. On the other hand many of the existing firms which were making normal profits or even small supernormal profits will find themselves in subnormal profits because of the fall in the price and will leave the industry as they are not covering the full opportunity cost.



Supply will fall as the firms leave and price tends to rise. This adjustment or entry and exit continues until all the existing firms reach long run equilibrium where  $MC=MR$  and  $AC=AR$ . It is the price at which all the costs including the opportunity cost of capital are being covered and there is no incentive for the firm to leave the industry or the other firms to enter the industry.

## MONOPOLY

A pure monopoly is an industry in which there is one supplier of a good or service or resource that has no close substitutes. In this case the firm will be an industry and the firm's demand curve will be the industry's demand curve.

The government definition is that a firm is said to be a monopoly when it supplies at least 25% of the market, provided that rest of the firms are sufficiently small. In most cities governmentally owned or regulated public utilities - gas electric and water companies - are all monopolies. The classic example of private, unregulated monopoly is the De Beers which effectively controls 80 to 90 percent of the world's diamond supply. Monopoly may also be geographic. A small town may have only one airline or railroad. The local bank, movie or bookstore may approximate a monopoly in a small, isolated community.

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**CHARACTERISTICS:**

1. **Single seller**: A pure monopolist is one firm industry. A single firm is the only producer of a specific product; the firm and the industry are synonymous.
2. **No close substitutes**: The product is unique in that there are no good or close substitutes. From the buyer's view point there is no reasonable alternatives.
3. **Price makers**: A monopolist is a price maker; the firm exercises considerable control over the price because it is responsible for the total quantity supplied
4. **Blocked entry**: A pure monopoly has no immediate competitors because there are barriers to entry. Technical, legal or other obstacles must exist to keep new competitors from coming in to industry, if monopoly is to persist.

**FIRMS' OBJECTIVES**

1. **Profit maximization**: The traditional objective of firms is profit maximization. It is usually assumed that firms seek to make as high profits as possible and that is achieved when marginal cost (the change in cost resulting from producing one more unit) is equal to marginal revenue (the change in revenue from selling one extra unit).

**There are several reasons why firms do not operate at the profit maximization level of output:**

- It is difficult to identify this output. The firm may simply work out the average total cost and then add on a standard profit margin in order to determine the selling price. This cost-plus pricing technique is unlikely to result in maximum profit.
- Short-term profit maximization may not be in the long-term interest of the firm, since firms with large market shares may wish to avoid the attention of government regulatory bodies, such as the Competition and Markets Authority in the UK and the US Justice Department.
- Large supernormal profit may attract new entrants into the industry, especially where there are low barriers to entry.

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- High profits may damage the relationship between the firm and its stakeholders, such as its consumers and the firm's workforce, as they may see senior managers and shareholders earning large returns.
  - Profit maximization may not appeal to the management, who may have different objectives to the shareholders - high profits might trigger action by the firm's rivals and the firm could become a target for a takeover bid. This is another example of the principal- agent problem.
2. **Profit satisficing:** Some studies of firm's activities suggest that rather than trying to maximize profits, the managers of some firms aim for a satisfactory level of profits which will keep shareholders happy.
  3. **Behavioral theories:** Behavioral theories argue that groups connected with a firm have a variety of objectives for a firm. At any one time one particular objective may be paramount. Whether this is the case and which objective will depend on the relative strength of the different groups who influence the firm's objectives.

The groups who have a direct interest in the success of a firm are sometimes referred to as stakeholders. They include not only managers and shareholders but also workers, consumers, the government, the local authority and environmental groups. Each group will have a minimum level of demands. For example consumers are likely to want a minimum level of quality and workers a minimum level of pay and job satisfaction. The power, objectives and activity of these groups can change over time.

4. **Revenue maximization:** It is an alternative theory of a firm's behavior, related to the principal-agent problem. The separation of management from ownership, especially in large firms, can result in a firm's objective changing. Senior managers tend to be more interested in increasing sales and maximizing revenue from sales.

Managerial salaries and bonuses are usually based on total revenue, not profits. Sales can be easily and regularly monitored; it is much more difficult to measure profits. In principle, production will continue to the point where  $MR = 0$ . This means that a firm with a revenue maximizing objective will produce a higher output than a firm with a profit maximizing objective. The firm may produce where  $MC > MR$  provided  $MR > 0$  since total revenue will increase

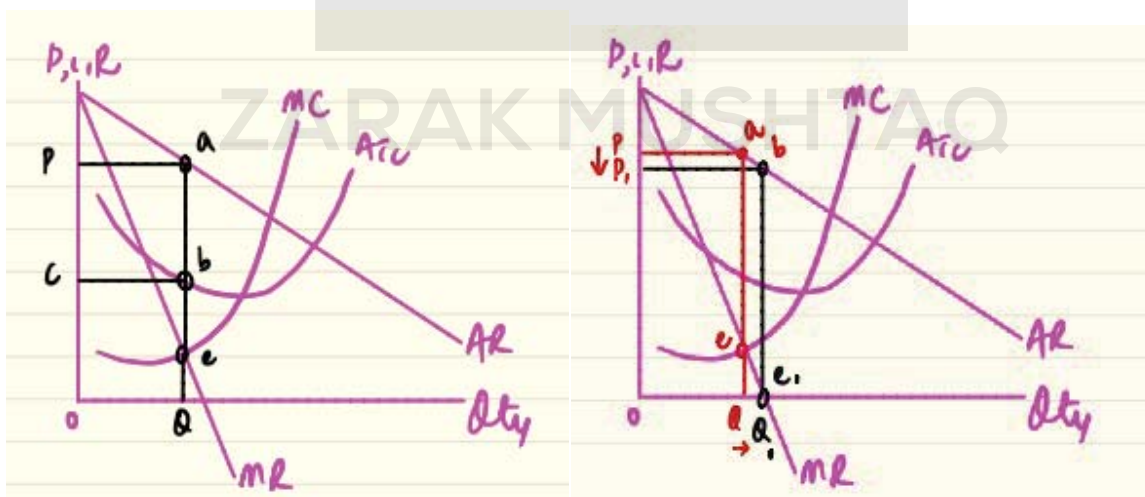
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5. **Sales maximization:** This objective is to maximize the volume of sales rather than the total revenue from sales. Sales maximization will lead to greater output than in the case of revenue maximization. With sales maximization, the firm would increase output up to the break-even output where the total revenue just covered the total cost. A higher output than this implies loss-making behavior. The only situation where loss making would be possible is where the firm could use the profit from one part of the firm to offset losses made elsewhere. This is referred to as cross subsidization. It could be that a state-owned firm, such as one running local bus services, uses the profits from city services to retain lower density routes that are loss-making. In this case, there are social objectives lying behind price and output decisions. The company might be instructed to keep prices down, to cover their ATC, or to make sufficient profit to be self-financing when it comes to new investment.

Sales can also be maximized where a firm follows an objective of growth maximization through increasing market share. The benefits of economies of scale can apply when a firm acquires or merges with a competitor.

The deliberate cutting of price to make a loss might be a strategy to deter new entrants into the market. Predatory pricing can also be used to force a rival firm out of the market. The legality of predatory pricing is questionable



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**The principal-agent problem**

In making decisions about the growth of firms, it is assumed that those who make the decisions have the authority to do so. In small firms, this is most likely to be the owner or partners who agree on how growth might or might not occur. They have the authority to act. In large firms, especially those with shareholders, this is not always the case. What can arise is a gulf in decision-making between the firm's management and ownership. This is known as the principal-agent problem.

The principal-agent problem occurs when one person (the agent) makes decisions on behalf of another person (the principal). The agent, through a day-to-day involvement in the firm, has more information than the principal. The problem is that the principal does not know how the agent will act; the principal is also not sure that the agent will act in the principal's best interests.

In the case of a firm's growth, the agent may have plans and a strategy that differ from those of the principal. The agents may decide to act in their own interest. If successful, the agents stand to gain prestige and enhance their own career development. The principal is not fully aware of what the growth plans involve. This is called the agency cost.

The principal-agent problem is an example of market failure that occurs because of information failure. The principals should be the ones in the best position to determine a firm's future growth. This is often not so and leads to a misallocation of resources.

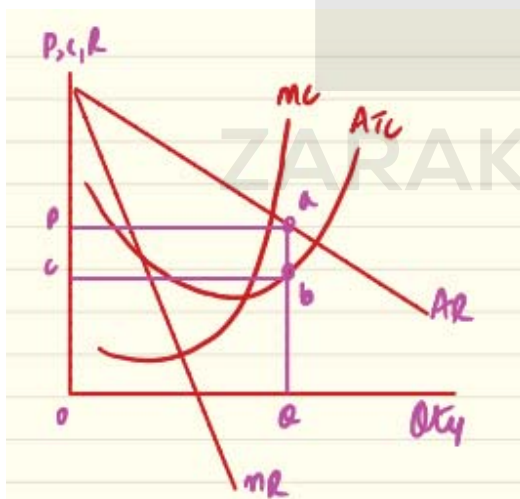
**CONTESTABLE MARKETS:** In recent years, economists have developed the theory of contestable markets. This theory argues that what is crucial in determining price and output is not whether the industry is actually a monopoly or competitive but whether there is the real threat of competition. If a monopoly is protected by high barriers to entry, say that it owns all the raw material, then it will be able to make supernormal profits with no fear of competition. If, however, another firm could take over from it with little difficulty, it will behave much like a competitive firm. The threat of competition has similar effect as the actual competition.

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For example, consider a **catering company** that is given permission by a factory to run its canteen. The catering company has a monopoly over the supply of food to the workers in that factory. If, however, it starts charging high price or provide a poor service, the factory could offer the running of the canteen to an alternative catering company. This threat may force the original company catering company to charge reasonable price and offer a good service. Entry is usually costly to a firm. It may have to build a plant, it may have to develop new versions of differentiated product or it may have to advertise heavily in order to call attention to its product. These and many other initial expenses are often called sunk costs of entry, which are defined as costs that a firm must incur to enter a market and cannot be recovered if the firm subsequently exists.

A market is perfectly contestable when the costs of entry and exit by the potential rivals are zero and when the costs of entry can be made very rapidly. In such cases the moment the possibility of earning supernormal profits occurs, new firms enter, thus driving the profits down to normal profits. The sheer threat of this to happen will ensure that the firm already in the market will (a) keep its prices down so that it just makes normal profits and (b) produces as efficiently as possible, taking advantage of any economies of scale and new technology. If the existing firm did not do this, entry would take place and potential competition would become actual competition.



If a market is contestable, there is downward pressure on price, because the existence of supernormal profits provides a signal for new firms to enter the

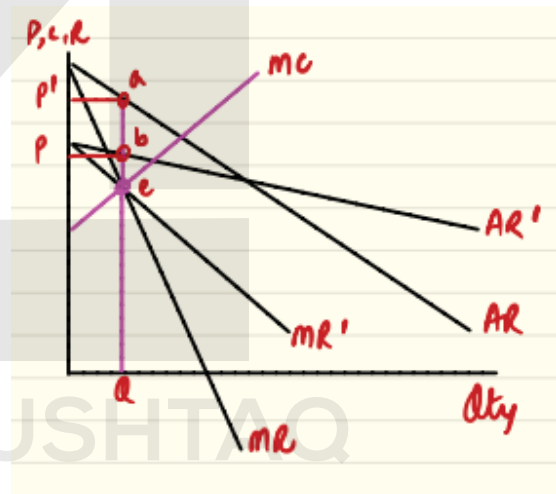
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market and if the existing monopolist is producing at too high a price then entrants can undercut the monopolist and some of the monopoly profit will be competed away. Normal profit equilibrium occurs when average revenue equals average total costs. From an economic welfare point of view, a lower price and higher output implies an improvement in consumer welfare (which could be illustrated by an increase in consumer surplus). When markets are genuinely contestable - we expect to see lower profit margins than when a monopoly operates without competition. Indeed the threat of competition may just as powerful an influence on the behavior of the existing firms in a market than the actual entry of new businesses. If a dominant firm in a contestable market knows that new suppliers may come in, this may be sufficient for them to charge a price closer to the level we might expect from a competitive market structure. If a market is contestable, industry structure and firm behavior is determined by the threat of competition - hit and run entry. The market will resemble perfect competition, regardless of the number of firms, since incumbents behave as if there were intense competition.

### No supply curve for monopoly: At

first glance we would suspect that the pure monopolist's MC curve would also be its supply curve. But this is not the case and pure monopoly has no supply curve. The reason is that there is no unique relationship between price and the quantity supplied. The price and amount supplied depend on the location of the demand curve and the MR curve.



Like the competitive firm, the monopolist equates MR and MC, but for the monopolist MR is less than price. Because the monopolist does not equate MC to price, it is possible for different demand conditions to bring about different profit maximizing prices for the same output, this represents price discrimination.

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**BARRIERS TO ENTRY:** The power of the monopolist depends on the effective barriers to entry which restrict the new firms to enter the industry. Some of these barriers are:

1. **ECONOMIES OF SCALE:** Modern technology in some industries is such that efficient, low cost production can be achieved only if producers are extremely large. Where EOS is very significant, a firm's LRAC will decline over a wide range of output. Small scale entrants cannot realize the profits necessary for survival and growth. If monopoly experiences substantial EOS, the industry may not be able to support more than one producer which is called as a natural monopoly. Most of the public utilities i.e. electric and gas companies, bus firms, local water and telephone companies are natural monopolies. It would be wasteful if a community had several firms supplying water and electricity.
2. **PATENTS:** The firm's monopoly position may be protected by patents which are issued to an inventor to protect them from having the invention copied for a period of time. Patent control figured prominently in the growth of modern day industrial giants such as General Motors, Xerox, Polaroid, and General Electric etc.
3. **OWNER SHIP OF ESSENTIAL RESOURCES:** A firm controlling a resource essential to the production process can prohibit the creation of rival firms. At one time the International Nickel Company of Canada controlled 90% of the world's known nickel reserves. Most of the world's known mines are owned by the De Beers Company of South Africa.
4. **TECHNOLOGY:** In modern capitalist states there are a number of industries which are dominated by a few giant firms. Where the existing firms are already on a vast scale, making use of expensive capital can enjoying extensive technical EOS, the barriers to entry of new firms are quite formidable. There is very little prospect of a new entrant starting on a small scale and growing larger by increasing his share of the market. The smaller producer will not be able to compete the larger scale producer because his average cost would be much greater than the larger scale producer.
5. **RESTRICTIVE PRACTICES:** Although in most countries they are now subject to legal control and many of them have been illegal, agreements

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between firms to restrict competition in an industry have been an important means of establishing monopoly situations. There are many ways in which the existing firms in an industry can combine to prevent entry of new firms. For example they might act collectively to withhold supplies from any wholesaler or retailer who stocks the goods of any new producer.

### What is Price Discrimination?

Price discrimination refers to a pricing strategy that charges consumers different prices for identical goods or services

#### Different Types of Price Discrimination

##### 1. First Degree Price Discrimination

Also known as perfect price discrimination, first-degree price discrimination involves charging consumers the maximum price that they are willing to pay for a good or service. Here, consumer surplus is entirely captured by the firm. In practice, a consumer's maximum willingness to pay is difficult to determine. Therefore, such a pricing strategy is rarely employed.

##### 2. Second Degree Price Discrimination

Second-degree price discrimination involves charging consumers a different price for the amount or quantity consumed. Examples include:

- A phone plan that charges a higher rate after a determined amount of minutes are used
- Reward cards that provide frequent shoppers with a discount on future products
- Quantity discounts for consumers that purchase a specified number of more of a certain good

##### 3. Third Degree Price Discrimination

Also known as group price discrimination, third-degree price discrimination involves charging different prices depending on a particular market segment or consumer group. It is commonly seen in the entertainment industry.

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For example, when an individual wants to see a movie, prices for the same screening are different depending on if you are a minor, adult, or senior.

### Primary Requirements for a Successful Price Discrimination

For a firm to employ this pricing strategy, there are certain conditions that must be met:

#### #1 Imperfect competition

The firm must be a price maker (i.e., operate in a market with imperfect competition). There must be a degree of monopoly power to be able to employ price discrimination. If the company is operating in a market with perfect competition, this pricing strategy would not be possible, as there would not be sufficient ability to influence prices.

#### #2 Prevention of resale

The firm must be able to prevent resale. In other words, consumers who already purchased a good or service at a lower price must not be able to re-sell it to other consumers who would've otherwise paid a higher price for the same good or service.

#### #3 Elasticity of demand

Consumer groups must demonstrate varying elasticities of demand (i.e., low-income individuals being more elastic to airplane tickets compared to business travelers). If consumers all show the same elasticity of demand, this pricing strategy will not work.

#### Example of Price Discrimination: Cineplex

The Canadian entertainment company, Cineplex, is a classic example of a firm using the price discrimination strategy. Depending on the age demographic, tickets for the same movie are sold at different prices. In addition, Cineplex charges different prices on different days (Tuesday being the cheapest and weekends being the most expensive).

|                         | Child<br>(3-13) | General<br>(14-64) | Senior<br>(65+) |
|-------------------------|-----------------|--------------------|-----------------|
| General                 | \$9.25          | \$13.25            | \$9.99          |
| 3D                      | \$12.25         | \$16.25            | \$12.99         |
| UltraAVX                | \$12.25         | \$16.25            | \$12.99         |
| UltraAVX<br>3D          | \$14.25         | \$18.25            | \$14.99         |
| D-BOX<br>UltraAVX       | \$20.25         | \$24.25            | \$20.99         |
| D-BOX<br>UltraAVX<br>3D | \$20.25         | \$24.25            | \$20.99         |

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**Advantages of Price Discrimination**The Firm

- **Profit maximization:** The firm is able to turn consumer surplus into producer surplus. In a first-degree price discrimination strategy, all consumer surplus is turned into producer surplus. It also ties into survivability, as smaller firms are able to better survive if they are able to offer different prices in times of greater and lower demand.
- **Economies of scale:** By charging different prices, sales volume is likely to increase. As a result, firms can benefit from increasing their production towards capacity and utilizing economies of scale.

The Consumer

- **Lower prices:** Although not all consumers are winners, consumers that are highly elastic may gain consumer surplus from the lower prices, due to price discrimination. For example, at a movie theatre, tickets for seniors and children are typically priced at a discount to adult tickets.

**Disadvantages of Price Discrimination**

- **Higher prices:** As indicated above, some consumers will face lower prices while others will face higher prices. Consumers that face higher prices (i.e., consumers who purchase airline tickets during peak season) are disadvantaged.
- **Reduction in consumer surplus:** The pricing strategy reduces consumer surplus and transfers money from consumers to producers, leading to inequality.

**ADVANTAGES of monopoly:** In the case of natural monopoly where large scale capital is required, average and marginal costs can fall significantly of the output is produced by one firm. Even in the case of monopolies, which are not natural monopolies, significant advantage may still be taken of EOS. The lower average cost may result in prices being lower than under other forms of market structures. Monopolies may spend significant sums on research and development as they are likely to have the finance to do so and the ability to protect the gains derived from

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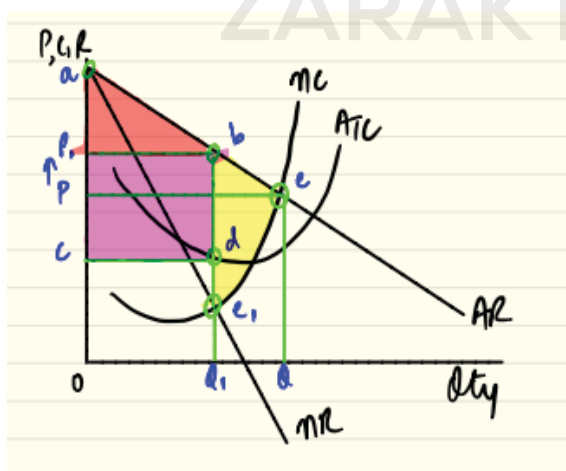
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any ideas developed. Some economists argue that innovations are more likely to occur under conditions of monopoly. This is because existing firms may bring new techniques and products into use to strengthen their barriers to entry and because firms outside the industry will try to innovate to get round the barriers to entry.

In addition monopoly in the home market might be necessary in order to obtain important EOS which, in turn, would lead to lower-priced exports. Firms which operate arguments to restrict competition between them might, as a consequence, collaborate in cost reducing research and development. An agreement to restrict competition might be necessary in order to ensure a domestic source of supply.

Another argument in favor of monopoly is that it provides greater stability of output and prices, a monopolist is likely to react to demand changes in a more effective manner, he supplies the total market and should be capable of estimating the true extent of market trends much more accurately than a small firm supplying a tiny part of the market. His adjustments of output, therefore, are likely to bring about an equilibrium situation fairly quickly.

**DISADVANTAGES:** A purely monopolistic firm will produce much less desirable results. The monopolist will find it profitable to sell a smaller output and to charge a higher price than would a competitive producer. The monopolist's profit maximizing output results in an under allocation of resources; the monopolists finds it profitable to restrict output and therefore employ fewer resources than are justified from society's point. **Neither allocative nor productive efficiency is achieved in monopolized markets.**



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**Income distribution:** In general, monopoly contributes to inequality in income distribution. Monopolists charge a higher price than would a purely competitive firm with same costs; monopolists in effect can levy private tax on consumers and obtain substantial profits. These monopolistic profits are not widely distributed because corporate stock ownership is largely concentrated in the hand of upper income groups. The owners of monopolistic enterprises tend to be enriched at the expense of the rest of the society.

**Producer sovereignty:** In contrast to those who believe monopolists will be innovative, some argue that a monopolist has little incentive to innovate, market control guarantees profit, with or without innovation, monopolists are also accused of retarding technical progress by restricting the entry of new forms. Under monopoly, while there may be range of products on offer there will be lack of choice of producers. The producers will be in stronger position than consumers' sovereignty likely to be low, consumer surplus is likely to be less under monopoly than under other market conditions.

### Natural monopoly

A natural monopoly is a market situation where a monopolist has overwhelming cost advantage. In theory, this can occur where a monopoly has sole ownership of a resource or where past ownership of capital resources means that it is difficult and extremely expensive for competitors to duplicate these factors of production. It would also be extremely wasteful to do so.

The natural monopoly case is one that has consistently been put forward in support of public ownership of services such as water, gas and electricity. It is also applicable to railways and inland waterways. In these cases, fixed costs are a very high percentage of total costs. As output increases, average fixed costs will decline, offering the prospect of substantial benefits to be gained from economies of scale. With a railway, for example, long-run average costs per train using the railway track will fall since the total costs are spread over an increased number of trains.

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## Market structures

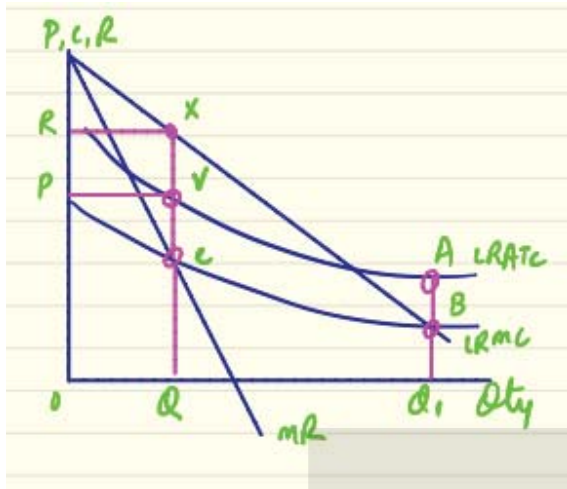


Diagram shows a falling long-run average cost curve, indicative of continuing economies of scale. A monopoly firm would set price at  $P$ , provide  $Q$  output and earn supernormal profit. If the monopoly behaved like a competitive firm, the equilibrium position would be where price equals long-run marginal cost - price would be lower, the quantity provided higher. This is a loss-making position. The problem is that this will only happen if the government is prepared to subsidize the monopoly on the grounds that what it is providing is an essential public service. The alternative is for the service to be provided by the public sector with losses funded from within the organization.

The natural monopoly case is particularly strong in the case of railways, canals, piped water supplies and electricity. There is little economic logic in having rival companies competing with each other, especially as with rail government subsidies are required to keep the system functioning at a social optimum level.

### MONOPOLISTIC COMPETITION

It is a market structure in which there are many firms competing but where each firm has some degree of market power i.e. each firm has some choice over what price to charge for its products.

**CHARACTERISTICS:** The defining characteristics of monopolistic competition are:

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**RELATIVELY LARGE NUMBER OF FIRMS:** Monopolistic competition does not require hundreds or thousands of firms as do pure competition but only a fairly large number. Several characteristics of monopolistic competition from the presence of relatively large numbers e.g. each firm has a comparatively small percentage of the total market, so each has limited control over the market price. A relatively large number of firms ensure that collusion (concerted action by firms to restrict output and charge monopoly price) is not possible. With numerous firms in the industry, there is no feeling of mutual interdependence among them; each firm determines its policies without considering reactions of rival firms. Rival's reactions can be ignored because the impact of one firm's action on each of its many rivals is so small that these rivals will have no reason to react.

**PRODUCT DIFFERENTIATION:** In contrast to pure competition, monopolistic competition has the fundamental feature of product differentiation. Purely competitive firms produce a standardized or homogenous product, monopolistically competitive producers turn out variations of a particular product. Because of product differentiation, economic rivalry typically takes the form of non-price competition (competition in terms of product quality, services to customers, location and accessibility, and advertising). Product differentiation may take the form of physical or qualitative differences in products themselves. Personal computers, for example in terms of hardware, capacity, software, and how user-friendly they are. Most cities have a variety of retail stores selling men's and women's clothing varying greatly in styling, materials and quality of work.

Services and conditions surrounding the sale of a product are forms of product differentiation. The snob appeal of a store, the courteousness and helpfulness of clerks, the firm's reputation for servicing or exchanging the products, and credit availability are all services aspects of product differentiation. Pizza restaurants compete on the basis of whether they offer delivery services.

Products may also be differentiated as to location and accessibility. Small mini groceries or convenience stores successfully compete with large super markets on the basis of location. Product differentiation may also arise from perceived differences created through advertising, the use of brand names and trade-marks,

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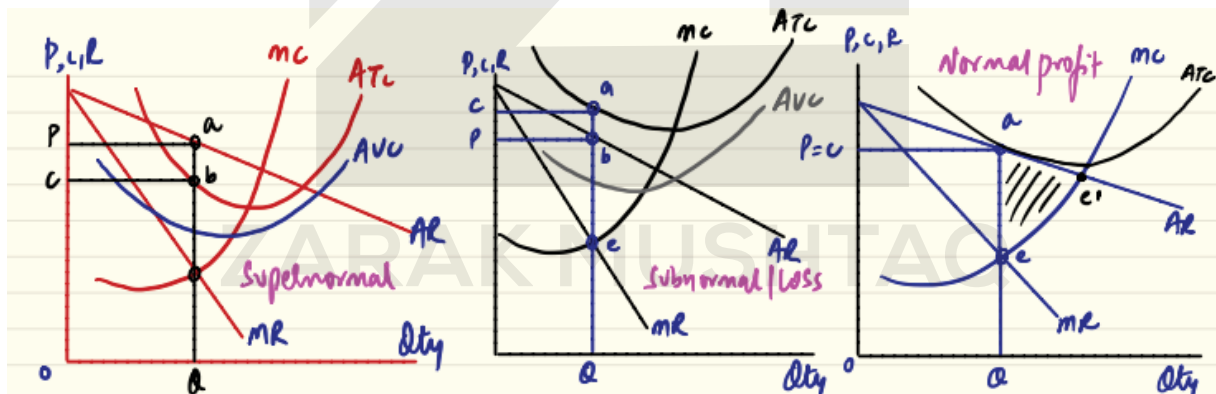
## Market structures

and packaging. E.g. a celebrity's name associated with jeans or perfume may enhance those products in the minds of buyers.

**EASY ENTRY:** Entry in to monopolistically competitive industries is relatively easy. The fact that monopolistically competitive producers are typically small sized firms suggests that EOS and capital requirements are few. On the other hand, compared with the pure competition, added financial barriers may result from the need to develop a product different from one's rivals and the obligation to advertise it. Some existing firms may hold patents on their products and copyrights on their brand names and trademarks, enhancing the difficulty and cost of successfully imitating them.

**THE FIRM'S DEMAND CURVE:** The demand curve faced by monopolistically competitive seller is highly elastic, but not perfectly elastic. It is much more elastic than the demand curve of a pure monopolist, because the monopolistically competitive seller faces many rivals producing close substitute goods.

**THE SHORT RUN EQUILIBRIUM: PROFITS OR LOSSES:** The firm will maximize its profits or minimize its loss in the short run by producing that output where  $MC=MR$ .



**THE LONG RUN:** In the long run a monopolistically competitive firm will earn only a normal profit. In the short run profits case, economic profits will attract new rivals, because it's easy to enter the industry. As new firms enter, the demand curve faced by typical firm will shift to the left i.e. demand will fall and the demand curve will become more elastic as each firm has a smaller share of the

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## Market structures

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total demand and now faces a larger number of close substitutes. This causes the economic profits to disappear.

When the industry suffers short run losses, some firms will exit in the long run. Faced with fewer substitutes and blessed with an expanded share of total demand, surviving firms will see their losses disappear and gradually lead to normal profits situation.

**MONOPOLISTIC COMPETITION AND ECONOMIC EFFICIENCY:** Economic efficiency requires the triple of price, marginal cost and the average cost. When price and marginal cost are equal, allocative efficiency will be achieved (the allocation of the right amount of resources to the product). When the price equals the minimum of average total costs, productive efficiency or the use of the least cost combination of the factors of production will result. Productive efficiency means consumers will enjoy the large volume of the product and the lowest price which least cost conditions allow. In monopolistically competitive markets excess capacity occurs and neither allocative nor productive efficiency is realized. Furthermore, in contrast to purely competitive firms, monopolistically competitive firms' are characterized by excess capacity, meaning that they produce somewhat short of the most efficient output.

### Advantages and disadvantages of monopolistic competition

#### Advantages

- Wide variety of differentiated products
- Prices likely to be lower and output higher than under monopoly
- High levels of competition encourage firms to keep costs down

#### Disadvantages

- Price higher and output lower than in perfect competition
- $\text{Price} > \text{MC}$  - Allocatively inefficient
- $\text{Price} > \text{minimum AC}$  - Productively inefficient. This also means firms have excess capacity. Because there is a large number of firms

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producing very similar products, they are not producing at their maximum capacity (the minimum point on the AC curve). Consumers pay for this through higher prices

- High levels of competition encourage firms to provide high-quality products
- Non price competition may be wasteful in terms of resources, e.g., advertising campaigns creating minor differences between products

**OLIGOPOLY**

An oligopoly is an industry that contains only a few competing firms. Each firm has enough market power to prevent its being a price taker. The number of competitors is small enough for each firm to realize that its competitors may respond to anything that it does and that it take such possible responses in to account. In other words, oligopolies are aware of the interdependence among the decisions made by the various firms in the industry and they engage in the type of rivalrous and strategic behavior. In this market structure the prices are typically administered and firms produce standardized or differentiated produces. Some of the common examples of oligopoly industries are oil, automobile, steel, air-line etc.

**WHY OLIGOPOLY OCCUR?**

1. **Economies of scale:** Perhaps the strongest reason that has been offered for the existence of oligopoly is EOS. When EOS occurs, the firms LRAC curve will be downward sloping as it produces more and more output. That is, ATC can be reduced by continuing to expand the scale of operation. Smaller firms will have a tendency in such a situation to be inefficient; their ATC will be greater than those incurred by a large firm and they will tend to go out of business. Historically in many of the industries that have become oligopolistic, it has been technical progress that has made EOS obtainable. In recent years new technologies can make smaller scales feasible, for

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example in printing and publishing. So technical EOS will not always be important. In fact it is often now the case that, where industries are heavily concentrated and strongly oligopolistic, the crucial factor is the marketing EOS which make it possible for large firm to advertise extensively.

In today's world the product development cost of one or few wide range of firms' product is very high. Of the firm sell large volumes of it products then the development cost can be spread over a large volume of the products sold and the unit cost will be very low. This can be covered in a relatively lesser period of time before some rival firms comes up with a new version of the product causing a fall in the demand of the existing product. Similarly other types of costs related to marketing and financing can also be covered relatively quickly.

2. **Mergers and takeovers:** Another reason that explains the development of oligopoly market structure is the increase in the size of the firms through mergers and takeovers. A firm that is larger both absolutely and relative to the market may have greater ability to control the market for, and the price of, its product does a smaller, more competitive producer. Also, the larger size which merger entails may give the firm the advantage of being a big buyer and permit it to demand and obtain lower price from input suppliers. Mergers allow companies to grow and indeed to expand overnight. However, sometimes a firm is reluctant to lose its independence and is faced not with a merger option but with a takeover bid from a rival. This process increases the size and market share of the survivors and the reduced competition allow them to earn larger profit margins.
3. **Capital required:** An important reason for the existence of a few firms in an industry is that the capital investment required to enter certain industries - the cost of obtaining necessary plant and equipment - is so great as to discourage entry. The cigarette, automobile, steel and petroleum refining industries are usually characterized by high capital requirements.

### DIFFERENT BEHAVIORS OF OLIGOPOLY MARKET

1. **Diversity:** Oligopoly encompasses a greater range and diversity of market situations than other market structures. It includes both; product

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differentiation and product standardization. It includes case where firms act in collusion and those where they act independently. It embodies situations in which barriers to entry are very strong and those in which they are not quite so strong. In short, the diversity of oligopoly precludes development of a simple market model which provides a general explanation of oligopolistic behavior.

2. **Interdependence:** Because there are only few firms under oligopoly, each firm will have to take account of the others. This means that are mutually dependent or interdependent. Each firm is affected by its rival's actions. If a firm changes the price or the amount of its advertising, the sales of its rival will be affected. The rivals may then respond by changing their price or advertising. No firm can afford to ignore the actions and reactions of other firms in the industry. It is impossible, therefore, to predict the effect on firm's sales of a change in its price without first making some assumptions about the reactions of other firms.

Oligopoly market structures may therefore have two distinct behaviors; the interdependence of firms may make them wish to collude with each other and maximize their joint profits. On the other hand; they will be tempted to compete with their rivals to gain a bigger share of industry profits for themselves.

In economics, a **concentration ratio** is a measure of the total output produced in an industry by a given number of firms in the industry. Concentration ratios are usually used to show the extent of market control of the largest firms in the industry and to illustrate the degree to which an industry is oligopolistic. For example, four firms concentration ratio is 25% which means that the largest four firms' market share totals 25%

**COLLUSIVE OLIGOPOLY (COOPERATIVE BEHAVIOR):** When firms agree to cooperate in order to restrict output and to raise prices, their behavior is called explicit collusion. Collusive behavior may occur with or without an actual agreement to collude. Where explicit agreement occur economist speak of overt or covert collusion, depending on whether the agreement is open or secret.

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1. **CARTEL:** a formal collusive agreement is called a cartel. It is an agreement between two or more firms to produce a combined monopoly output by allocating output quota to each firm. In this situation the firms behave as if they were a single seller in order to maximize their joint profits and eliminate competition among themselves. When a competitive industry is cartelized, the total output of the industry falls and the price increases. An effective cartel requires the existence of barriers to entry and that the firms must follow the agreement by producing the output which maximizes the industry's joint profits. An entry of new firms in to the industry will lead to increase in supply causing output to increase more than the profit maximizing level and hence the industry profits tend to fall. Similarly, each individual cartel member can increase its profits by violating the output restrictions, provided that the other members do not violate.

### Conditions for an effective cartel

There are a number of conditions necessary for a cartel to operate successfully. These include the following:

- A limited number of firms in the industry, all of whom are members
- Very high concentration ratios. Studies suggest that two-thirds of cartels are in industries with a four firm concentration ratio in excess of 75 per cent
- Control over the source of supply. For example, it has been estimated that OPEC control 50 per cent of global oil supplies and 90 per cent of proven oil reserves
- An ability on the part of the members to control price and restrict output
- Similar cost structures for all firms. If one or more companies has significantly lower costs than the others they may have an incentive to increase their output and reduce price in order to increase their profits
- High barriers to entry to prevent new firms entering the industry
- All firms obey the rules of the cartel and there are measures in place to ensure compliance with the rules possibly involving sanctions

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- A stable market - cartel agreements are likely to be vulnerable in times of recession when total demand in the economy is falling. This is a particular problem if the product is One with a high-income elasticity of demand
- The firms' produce very similar or identical products which inspire little consumer loyalty, such as industrial components. This tends to take agreements On price easier to establish.

2. **PRICE LEADERSHIP:** It is a common feature of oligopolistic markets. All firms in the market accept the price that is set by the leading firm, which is often the firm with the largest market share or is the brand leader. They then alter their own prices in line with those of the leader. It is seen as a way of avoiding price competition yet maximizing total profits for all firms while allowing various forms of non-price competition to prevail. A good example in many countries is petroleum retailing, where the market leader is a well-known multinational company. Other examples are in the markets for certain types of retail goods or branded fast food.

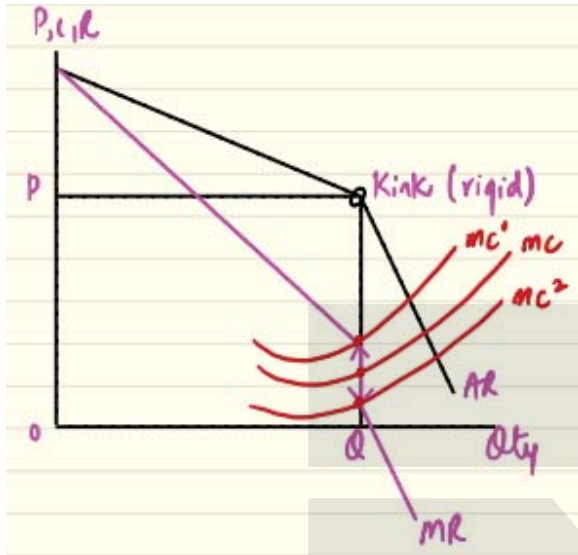
When the leading firm announces a price increase, all others quickly follow, sometimes in just a matter of hours. It is the same when a price fall is announced; smaller rivals are forced to follow the fall in price to retain market share. However, the downside is that since their costs are higher, matching a price decrease could result in sustained losses and the eventual exit of smaller rivals from the market.

**NON-COLLUSIVE OLIGOPOLY (NON-COOPERATIVE BEHAVIOR):** The kink demand curve is an operational tool for the determination of the equilibrium in the oligopolistic markets. The kink demand curve is made out of the firms' own demand curve and the market demand curve. The firm's own demand curve is highly elastic because of the availability of very close substitutes and the industry's demand curve are highly inelastic because there is no close substitute for the industry's product. The upper section of the kinked demand curve has a higher elasticity because it is the part of the firm's own demand curve and the lower part has a lower elasticity because it is part of the industry's demand curve. Due to the kink

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in the demand curve of the oligopolies, MR curve is discontinuous at the level of the output corresponding to the kink



**Game theory** involves the analysis of the strategies adopted by rational players (individuals, groups, firms or even nations) in situations when each player must make decisions in the knowledge that the final outcome (payoff) will depend on the reactions of other players and where the first player knows the other is thinking in the same way. Game theory has been applied to an enormous range of activities and aspects of human behavior including wars, chess, driving, choosing a marital partner and selling a house. It has applications across a range of disciplines including both the physical and social sciences.

All games, however, have the common features that they are played out in an environment of uncertainty, interdependence and often conflicts of interest amongst the players. Here we concentrate on relatively straightforward two-player zero-sum games where one player's gain must equal the other player's loss.

**The prisoner's dilemma in oligopolistic markets and the two-player matrix:** The most famous and often quoted example of game theory strategy is the prisoner's dilemma. In this example, two individuals, Adam and Ben, have been arrested on suspicion of having committed a very serious crime, which carries a maximum prison sentence of ten years.

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Adam and Ben are being detained in separate parts of the prison and have no way of communicating with one another. The chief of police meets with each in turn and speaks to them in the same way:

"I have sufficient evidence to send both of you to prison for three years if neither of you confesses. However, if you alone are prepared to confess to the crime you will receive a light sentence of one year and your partner will receive a prison sentence of ten years.

If both of you confess, you will both receive a sentence of five years". What should Adam and Ben do? What is the best strategy for each in these circumstances?

|     |             |                                                               |                                                               |
|-----|-------------|---------------------------------------------------------------|---------------------------------------------------------------|
|     |             | Adam                                                          |                                                               |
|     |             | Not Confess                                                   | Confess                                                       |
| Ben | Not confess | <p>A</p> <p>Adam</p> <p>3 years</p> <p>Ben</p> <p>3 Years</p> | <p>B</p> <p>Adam</p> <p>1 Year</p> <p>Ben</p> <p>10 Years</p> |
|     | Confess     | <p>C</p> <p>Adam</p> <p>10 Years</p> <p>Ben</p> <p>1 Year</p> | <p>D</p> <p>Adam</p> <p>5 Years</p> <p>Ben</p> <p>5 Years</p> |

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Let us begin with Adam. He will reason as follows.

It is best that I confess because in the best case scenario I will receive a light sentence of one year if Ben does not confess and in the worst case scenario I will receive a sentence of five years rather than ten years in prison if Ben also confesses.

Ben will follow the same path of reasoning and also conclude that the best strategy is to confess.

The final result is that both Adam and Ben will confess and receive a sentence of five years each as illustrated in quadrant D of the matrix. Clearly this is not the best outcome (pay-off) for Adam and Ben. The best outcome would be for them to co-operate or collude and both agree to not confess as illustrated in quadrant A, which would give them a lighter sentence of three years. However, in the absence of any ability to communicate with one another this is unlikely, and even if they could communicate with one another they would each have to trust in the other's willingness to stick to the deal.

### **Game theory and oligopoly**

The application of game theory to the behavior of firms in oligopolistic markets is most closely associated with the work of the Nobel Prize winner in Economics for 1994, John Nash.

Let us take the example of a duopoly in the low-cost airline market- In this case two firms, Bat flight and Swing Airlines, are the players; the game takes place in the market and the pay-off is the expected final profit for each firm resulting from their respective strategies.

Currently both airlines are charging the same fare, \$100 return on a particular route. We assume that they are not prepared to co-operate with one another and are independently considering the impact on their profits if they reduce fares on the route to \$90.

In deciding whether to leave fares at \$100 or lower them to \$90, each airline will need to take into account the likely response of the other airline to the decision it

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makes. The pay-off matrix below (Figure 7.44) shows the profits made (in \$millions) by the two firms as a result of leaving fares at \$100 and reducing them to \$90.

Current profits for each firm are indicated in quadrant A. If we begin with Bat flight and assume that it decides to adopt a conservative approach, anticipating that whatever strategy Bat flight chooses, Swing will adopt a counter strategy which is best for itself but worst for Bat flight, then under these circumstances Bat flight should reduce its fares.

|                |                      | Batflight                                          |                                                    |
|----------------|----------------------|----------------------------------------------------|----------------------------------------------------|
|                |                      | Keep fares at \$100                                | Reduce fares to \$90                               |
| Swing Airlines | Keep fares at \$100  | <p>A</p> <p>Batflight \$30m</p> <p>Swing \$30m</p> | <p>B</p> <p>Batflight \$35m</p> <p>Swing \$15m</p> |
|                | Reduce fares to \$90 | <p>C</p> <p>Batflight \$15m</p> <p>Swing \$35m</p> | <p>D</p> <p>Batflight \$20m</p> <p>Swing \$20m</p> |

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**Maximin strategy:** The reason is this. If Bat flight reduces its fares to \$90 and Swing leaves its fares unchanged then Bat flight's profits will rise to \$35 million as shown in quadrant B. If on the other hand Swing responds by reducing its fare to \$90 also, then the worst that can happen to Bat flight's profits is that they will be reduced to \$20 million, which is obviously better than the reduction in profits to \$15 million if Bat flight had left its fares at \$100. This particular conservative strategy is referred to as the maximum strategy.

**Maximax strategy:** An alternative approach that Bat flight might take is a more optimistic, if riskier, strategy known as the maximax strategy which assumes that Swing will respond to any move by Bat flight which leads to the most advantageous outcome in terms of profits for Bat flight. In this case Bat flight would assume that Swing would leave its price unchanged whatever move Bat flight made.

However, this maximax strategy would still give the same result. It is still best for Bat flight to reduce its fares to \$90 as this would result in the position shown in quadrant B with Bat flight's profits rising to \$35 million.

As with the prisoner's dilemma example, the other player in this game, Swing, is faced with the same choice of strategies as Bat flight and so if it plays the game rationally it will come to the same conclusion and will decide its best strategy is to reduce its fares whatever Bat flight chooses to do.

**Dominant strategy:** Whether the players adopt the maximin or maximax strategies, the result is the same - both should reduce their fares and so reducing fares is a dominant strategy in that it is the best one for each player irrespective of that adopted by the other.

**Nash Equilibrium:** The final position in this game is shown in quadrant D with both airlines reducing their fares and achieving profits of \$20 million. This position is known as a Nash equilibrium it that it is the best for each airline given the circumstances in which each finds itself taking into account the likely decisions of the other, and as such neither airline will have any incentive to change its strategy unless the other does.

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It is important to note that while this final position represents equilibrium by acting independently in a non-co-operative fashion, the airlines have failed to achieve the best possible outcome for themselves. This is shown in quadrant A with both firms leaving their fares at \$100 and making profits of \$30 million.

**Collusion:** However, if as is likely in reality, this is not a single-move game but one which is repeated over time, it is likely that the airlines will come to realise that their mutual interests are best served by co-operating or colluding either tacitly or through a formal cartel.

However, as the pay-off matrix shows, there may still be incentives for one of the players to cheat and undercut the other in order to increase its own profits.

For collusion to be effective, therefore, there must be clear punishments or penalties for cheating that can be imposed for this together with a willingness and ability to implement them. For example, one reason for the OPEC cartel members maintaining their quotas once they have been agreed is that all members are aware that Saudi Arabia is in a position to flood the market with oil thereby significantly reducing its price which would penalize all members. Saudi Arabia actually took this action in 2020.

One strategy which may prove effective in ensuring collusion in repetitive games is a "tit for tat" strategy. In this strategy a firm co-operates in the first round of moves, but makes it clear that it will only maintain this stance if all the other firms continue to co-operate. If other firms cheat, then in the next round of moves the first firm will mimic the actions of the other firms in the previous round and cheat. This immediately punishes the cheats and ensures there are no long-term gains from cheating. This strategy has the advantage of providing a clear punishment without undermining the collusive agreement because the firm will only cheat if others do.

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